

LIQI SHI

Waterloo, ON

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PROFESSIONAL SUMMARY

Senior Network & Telecommunications Core Architect with 30 years of industry and research experience spanning 4G LTE/5G Cloud Core architectures, deep packet inspection (DPI), policy control, traffic management, and software development. Proven track record in core deployments and solving mission-critical network escalations. Holds a Ph.D. in Electrical & Computer Engineering with 11 published peer-reviewed papers in telcom and computer networks research.

TECHNICAL SKILLS

- **Mobile Core Architectures (4G/5G):** UDM/UDR, PCF, AMF, SMF, UPF, DPI/Heuristics, QCI/QFI Congestion Management, Ericsson EEA, CGNAT.
- **Traffic Management & QoE:** Sandvine DPI solutions, Cisco/Nokia QCI/QFI, Satellite Traffic Management, IPTV & VoIP QoE Optimization.
- **Protocols & Signaling:** Radius/Diameter, VoIP, RTCP, TCP/IP, 3GPP, Application layer, Video Streaming Studies.
- **Systems, Automation & Tools:** C/C++, Linux, Python, Bash Scripting, SQL (Vertica), VxWorks, OPNET, Wireshark, Kubernetes, Azure DevOps.

PROFESSIONAL EXPERIENCE

Senior Network & Core Architect

Xplore Inc. *Markham, Sept 2016 - Sept 2026*

Led end to end traffic management strategies across access, transport, and core to manage congestions and enhance customer QoE.

Key Technical Engineering Projects:

- **Ericsson Core Deployment & Testing (Fixed Wireless):** Served as lead technical contributor in end-to-end deployment, verification, and testing of **Ericsson UDM/UDR** and **PCF policies**. Integrated **DPI/heuristics engines** to enforce **QCI/QFI congestion management**. Worked on **Ericsson EEA** and **CGNAT** deployment and testing.
- **Congestion Management (Fixed Wireless & Satellite Networks):** Engineered application-layer QoE traffic scheduling by integrating classification data from DPI

engines with downstream schedulers. Implemented **Sandvine Fairshare**, **Sandvine DPI with Cisco/Nokia QCI/QFI**, and **Sandvine DPI with Satellite Traffic Schedulers**.

- **RADIUS Authentication & Accounting Integration:** Designed and deployed robust RADIUS architecture connecting network core functions to centralized RADIUS servers for secure authentication, session accounting, IP address management, and historical tracking.
- **Traffic Management & QoE Monitoring for IPTV & VoIP:** Architected end-to-end traffic management and real-time Quality of Experience (QoE) monitoring solutions tailored for real-time voice (VoIP) and streaming media (IPTV) traffic.
- **Network Performance & Analytics Studies:** Conducted deep-dive network performance studies using SQL data pipelines and packet analysis to analyze subscriber traffic flows, optimize connections, and reduce protocol latency.

Software Engineer on DPI Solutions

Sandvine Inc. *ON May 2011 – Sep 2016*

- Served as engineering prime for escalated, mission-critical customer issues passed from technical support requiring deep-dive technical diagnosis and resolution.
- Collaborated closely with technical support, solution architects, and subject matter experts to identify root causes and deliver targeted fixes for complex network edge cases.
- Diagnosed networking bugs and isolated protocol/packet flow issues for Sandvine Policy Traffic Switch (PTS) products.
- Developed and delivered hot patches and software workarounds for identified defects to restore service availability in live carrier environments.

Software Engineer / Project Lead for Telcom Protocol Stack

Taicom Data Systems Inc. *Beijing, Jul 2000 – Feb 2004*

- Led an engineering team of 7 in designing protocols and software for an Always On / Dynamic ISDN (AODI) terminal adaptor.
- Designed and developed network protocols including PPP, ISDN, TCP/IP, and X.25 using VxWorks on MPC850/860 and ARM hardware platforms.
- Developed low-level hardware drivers and peripheral interfaces for telecommunication products.
- Conducted protocol verification and compliance testing using hardware protocol analyzers (HP / IDACOM PT500) against live telecommunication networks.

Software Engineer / Network Analyst

Malan Telecom XinJiang, *Jul 1996 – Jul 2000*

- Designed and built a functional simulator for the EWSD telecommunication switching system.

- Developed software tools to monitor live running status, performance, and fault states of telecommunications networks.
- Designed and deployed an operational X.25 data network.
- Analyzed traffic patterns and performance metrics for voice and data telecommunication networks.

RESEARCH & ACADEMIC EXPERIENCE

M.Sc. and Ph.D. Researcher / Graduate Assistant

University of Calgary | *Calgary, AB Jan 2005 – May 2011 (6 yrs 5 mos)*

- Proposed and evaluated a Quality of Service (QoS) constrained Ad Hoc On-Demand Distance Vector (AODV) routing protocol for tactical MANETs using OPNET (C/C++); presented outcomes directly to General Dynamics Canada.
- Designed a novel cross-layer optimization framework for Wireless Sensor Networks (WSNs) integrating Physical, MAC, and Network layers.
- Published 11 peer-reviewed papers in premier IEEE/ACM journals and international conference proceedings focused on mobile computing and wireless communications.
- Served on the Technical Program Committee (TPC) for IEEE CCNC (2009, 2010, 2011) and reviewed papers for major telecommunication journals.
- Served as Teaching Assistant for *ENEL529 (Wireless Communication Systems)* and *ENEL471 (Analog Communications)*.

EDUCATION & CERTIFICATIONS

- **Ph.D. in Electrical & Computer Engineering** | University of Calgary, Canada
- **M.Sc. in Electrical & Computer Engineering** | University of Calgary, Canada
- **B.Sc. / B.Eng. in Telecommunications / Electrical Engineering**
- **Vendor Training:** Ericsson 5G Cloud Core Architecture & Protocols